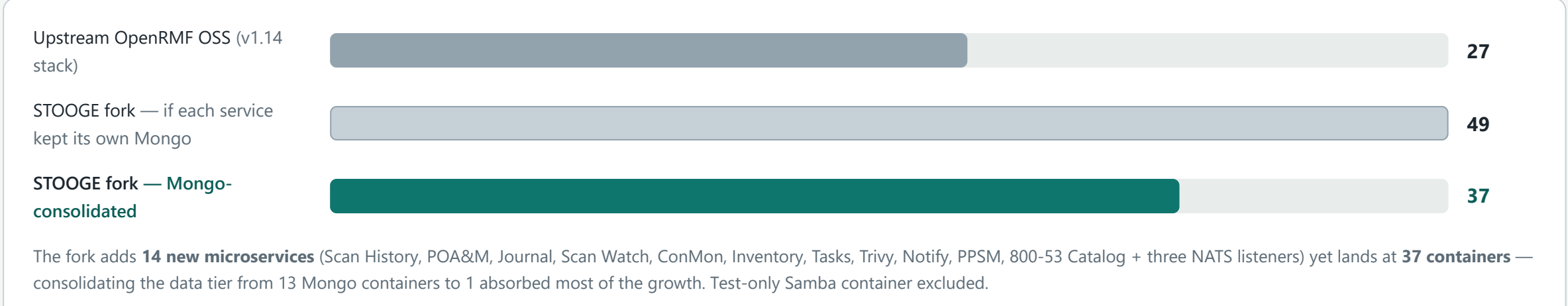


Architecture & Communication Ports / Protocols

Every box is a container on the internal `openrmf` Docker network. NGINX is the single host-exposed entry point; the NATS bus is the service backbone; every service owns exactly one **logical database** on a single consolidated MongoDB. All fork-built .NET services run on the **UBI8 (RHEL 8.10) base** with the DoD CA bundle. Teal marks fork additions/modifications.

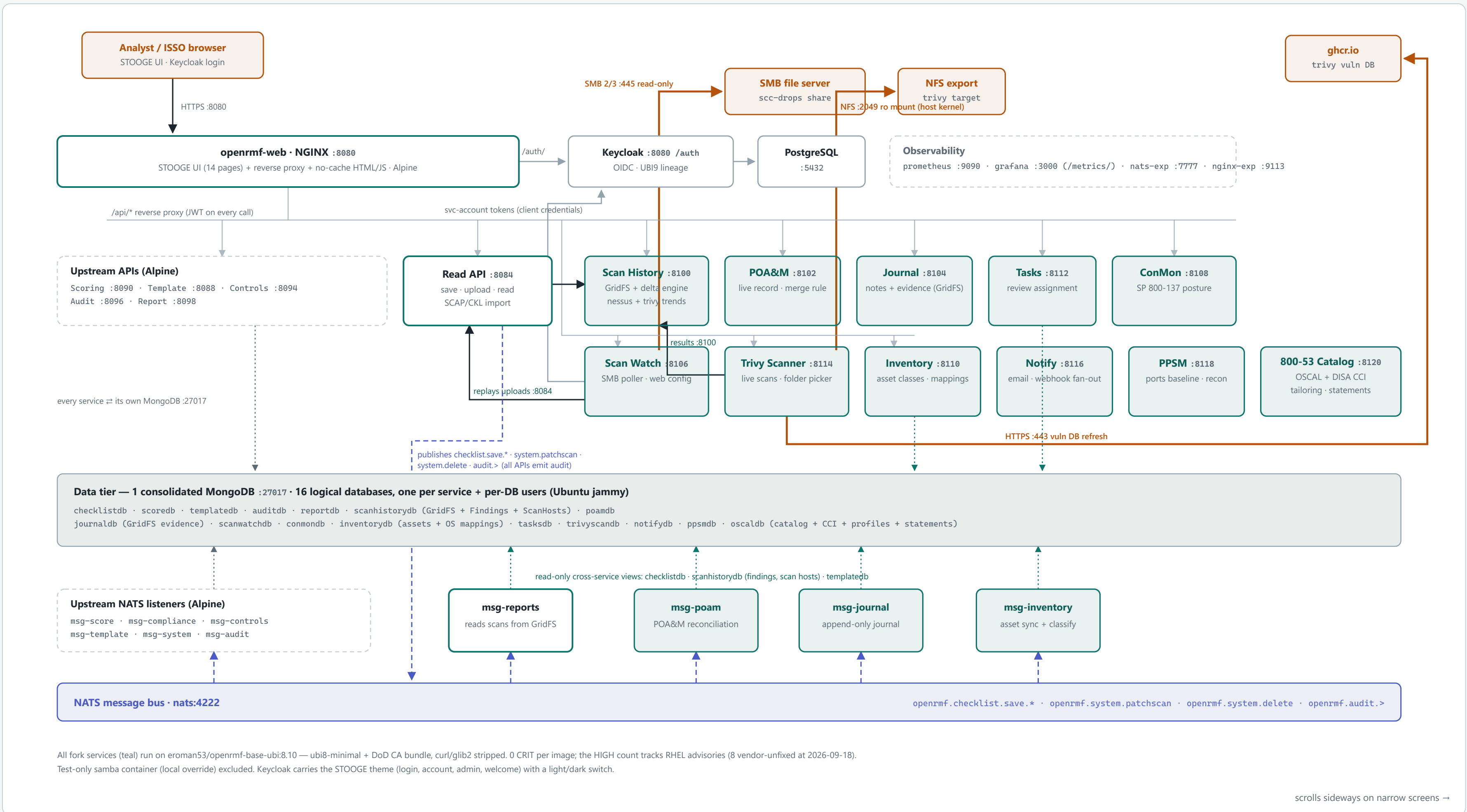
37 containers · 17 HTTP services · 10 NATS listeners · 1 Mongo container · 16 logical DBs · 1 host-exposed port (8080) · UBI8 fork services

Container footprint vs. upstream OpenRMF



1 · Container architecture

□ upstream (Alpine) □ fork-modified (UBI8) □ fork-new (UBI8) □ external | — HTTP / REST - - - NATS pub/sub MongoDB — SMB / NFS / registry (boundary)



All fork services (teal) run on eroman53/openrmf-base-ubi8.10 — ubi8-minimal + DoD CA bundle, curl/glib2 stripped, 0 CRIT per image; the HIGH count tracks RHEL advisories (8 vendor-unfixed at 2026-09-18). Test-only samba container (local override) excluded. Keycloak carries the STOOGE theme (login, account, admin, welcome) with a light/dark switch.

scrolls sideways on narrow screens →

2 · Communication ports & protocols (PPSM view)

INBOUND (HOST BOUNDARY)			
1 port			
8080/tcp HTTP — the only host-published port; UI, all APIs, and OIDC ride through NGINX. TLS on 8443 is provisioned (commented) in the config.			
OUTBOUND (HOST BOUNDARY)			
3 flows			
445/tcp SMB (scan drops, read-only) · 2049/tcp NFS (trivy target, read-only; +111 for v3) · 443/tcp HTTPS to ghcr.io (Trivy DB, air-gap: pre-seed the cache volume)			
INTERNAL ONLY			
everything else			
All service, database, bus, auth, and metrics traffic stays on the internal <code>openrmf</code> Docker network — nothing below the gateway is host-published.			

SOURCE	DESTINATION	PORT	PROTOCOL / SERVICE	PURPOSE
BOUNDARY-CROSSING				
Analyst / ISSO browser	openrmf-web (NGINX)	8080/tcp	HTTP (TLS 8443 available)	UI pages, all /api/* calls, OIDC login redirects
openrmf-svc-scanwatch	SMB file server	445/tcp	SMB 2/3	Polls the scan drop share (read-only account)
Docker host kernel	NFS server	2049/tcp (+111 v3)	NFS v4 (v3 opt.)	Read-only mount of the Trivy scan target
openrmf-svc-trivy	ghcr.io	443/tcp	HTTPS / OCI	Trivy vulnerability DB refresh (cached in volume; s
GATEWAY (NGINX REVERSE PROXY)				
NGINX	Keycloak	8080/tcp	HTTP / OIDC	/auth/ — login, tokens, JWKS
NGINX	Grafana	3000/tcp	HTTP	/metrics/ dashboards
NGINX	Read API	8084/tcp	HTTP REST	/api/read/ save - upload - read
NGINX	Template / Scoring / Controls / Audit / Report APIs	8088 · 8090 · 8094 · 8096 · 8098/tcp	HTTP REST	Upstream API routes
NGINX	Scan History / POA&M / Journal / Scan Watch / ConMon / Inventory / Tasks / Trivy Scanner / Notify / PPSM / 800-53 Catalog	8100 · 8102 · 8104 · 8106 · 8108 · 8110 · 8112 · 8114 · 8116 · 8118 · 8120/tcp	HTTP REST	Fork service routes (/api/scanhistory/ · /api/poam/ · /api/
IDENTITY & AUTH				
Every API / service	Keycloak	8080/tcp	HTTP / OIDC	Issuer discovery + JWKS for JWT validation on ew
Scan Watch · Trivy Scanner	Keycloak	8080/tcp	HTTP / OIDC token	Client-credentials tokens (service account openrmf
Keycloak	PostgreSQL	5432/tcp	PostgreSQL	Realm / user store
SERVICE-TO-SERVICE HTTP				
Read API	Scan History API	8100/tcp	HTTP REST	Streams .nessus uploads into GridFS + delta engin
Scan Watch	Read API	8084/tcp	HTTP REST	Replays dropped SCC XCCDF / CKL / Nessus files
Trivy Scanner	Scan History API	8100/tcp	HTTP REST	Uploads Trivy JSON as scanType=trivy snapshot
MESSAGING BACKBONE				
Read API (+ every API for audit events)	NATS	4222/tcp	NATS pub	openrmf.checkList.save.* · system.patc
10 msg listeners (score · compliance · controls · template · system · audit · reports · poam · journal · inventory)	NATS	4222/tcp	NATS sub	Event-driven reconciliation, journaling, scoring, re
DATA TIER				
Each service	Consolidated MongoDB — own logical DB + user	27017/tcp	MongoDB wire	One database per service, isolated by per-DB cre
msg-poam · msg-reports · msg-journal · msg-inventory · ConMon · Inventory API	checklistdb · scanhistorydb · templatedb · poamdb	27017/tcp	MongoDB wire	Deliberate read-only cross-service views (findings
OBSERVABILITY				
Prometheus	Service /metrics endpoints · NATS exporter · NGINX exporter	service ports · 7777 · 9113/tcp	HTTP scrape	Metrics collection across every tier
NATS exporter	NATS monitoring	8222/tcp	HTTP	varz / connz / subz
Grafana	Prometheus	9090/tcp	HTTP / PromQL	Dashboards (served to users via NGINX /metric

Sources: openrmf-docs/scripts/docker-compose.yml + nginx.conf as of the MongoDB consolidation (2026-07-20) on top of the UBI8 rebase (2026-07-18), refreshed 2026-09-18 for the Notify, PPSM and 800-53 Catalog services. Ports verified against the live nginx upstream config and the running containers; fork images scan 0 CRITICAL, with 8 vendor-unfixed HIGH inherited from the RHEL base. The PPSM module this matrix was drafted for is now built (8118).